AMP Challenge: A Benchmark for Generative AI in Antimicrobial Peptide Design

Ewa Szczurek[*] Paulina Szymczak Rasmus Møller-Larsen
Bruno Puczko-Szymański Marcelo Der Torossian Torres
Cesar de la Fuente-Nunez
ewa.szczurek@helmholtz-munich.de

Abstract

Antimicrobial peptides (AMPs) are a promising class of therapeutics against antibiotic-resistant infections, and generative AI has recently emerged as a powerful tool for their design. Yet most proposed methods are not validated in the wet lab. Experimental testing is costly, and the few studies that do report it use incompatible training data, bacterial strains, assay conditions, curation pipelines, and success criteria. Cross-study comparison is therefore impossible, and the field has no shared signal on which generative approaches actually work. This raises the central question that AMP Challenge is built to answer: which generative modeling strategies, computational metrics, and candidate-selection rules predict experimental antimicrobial efficacy? AMP Challenge is the first standardized benchmarking competition for generative AI applied to AMP design, in which computational predictions are directly falsified by wet-lab outcomes, and where progress claims in generative AI can be evaluated against reality rather than against other models. In the (i) computational phase, all submissions are scored under a fixed pipeline, and up to 20 qualifying teams advance. In the (ii) experimental phase, 25 peptides per team (up to 500 peptides total) are synthesized under a single vendor workflow. Minimum inhibitory concentration (MIC) and half-maximal hemolytic concentration (HC50) assays are performed in a single laboratory across 20 clinically relevant Gram-negative and Gram-positive strains, including 8 multi-drug resistant (MDR) isolates. Teams compete across five categories: broad-spectrum, Gram-positive, Gram-negative, MDR, and selectivity. Deliverables include an open benchmark dataset pairing computational metrics with standardized experimental MIC/HC50 data, a public model repository, and a jointly authored benchmarking paper. The benchmark will remain open after the competition closes, enabling continued evaluation of new methods against the same experimental ground truth. By releasing both successful and unsuccessful peptides, the competition directly counters the positive publication bias that has shaped the field to date.

Keywords Benchmark, Generative AI, Antimicrobial Peptide Design.

1 Competition description

1.1 Background and impact

Antimicrobial resistance (AMR) is one of the most pressing global health threats of the 21st century, contributing to making minor infections and routine surgeries life-threatening again. Antimicrobial peptides (AMPs) are a class of short, mostly cationic sequences produced across all domains of life

[*]Lead organizer.

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as part of the innate immune system. Because they act through diverse membrane-disruptive and intracellular mechanisms, AMPs can evade many of the resistance pathways that have eroded the utility of conventional antibiotics, and they are widely regarded as one of the most promising classes of next-generation therapeutics [25].

Designing effective AMPs is, however, a formidable combinatorial problem: even restricted to the 20 proteinogenic amino acids and a synthesis-friendly length of 8–50 residues, the sequence space exceeds 10^{10} candidates, the overwhelming majority of which are inactive or toxic. This setting is a natural fit for modern generative AI: language models [9, 14], diffusion models [11, 22, 27, 31], autoencoders [3, 24, 32, 36], generative adversarial networks (GANs) [2, 28, 35], GFlowNets [7], reinforcement learning [19], and Bayesian optimization [7] have all been applied to AMP design over the past five years.

Yet the field faces a reproducibility problem. Different groups train on different datasets, evaluate on different bacterial strains, use different assay conditions, apply different amounts of human *post hoc* curation, and use different definitions of what counts as a "hit". As a result, it is frequently unclear whether a reported success reflects the quality of the generative model itself, the quality of the manual selection process, or simply the choice of validation panel. Head-to-head comparisons under matched conditions are essentially absent [25].

AMP Challenge is designed to close this gap. It proposes a common benchmark with a shared experimental pipeline and transparent reporting requirements. Every submitted peptide that advances to validation is synthesized under a single vendor workflow and tested in a single laboratory against the same 20-strain panel, under the same protocol, blinded to team identity. The scientific question the competition answers is concrete:

> *Which generative modeling strategies, computational metrics, and candidate-selection rules predict experimental antimicrobial efficacy?*

The anticipated impact is threefold. Scientifically, the competition will produce the first controlled comparison of generative approaches for AMPs, together with an open dataset of 500 peptides with paired computational scores and experimental MIC/HC50 measurements across 20 strains–a uniquely valuable resource that reduces publication bias by releasing negative as well as positive results. Methodologically, it will establish a reusable protocol and evaluation framework that can serve as a template for future benchmarks in biological sequence design. Societally, by accelerating the principled development of AMP generators, the competition contributes directly to the long-term goal of new antimicrobials against MDR pathogens.

The problem is highly relevant to the NeurIPS community: generative modeling of biological sequences is an active area at the intersection of machine learning, computational biology, and drug discovery. We estimate 20–40 qualifying team submissions from ML-for-drug-discovery groups in academia and industry, protein language modeling labs, and computational chemistry groups.

1.2 Novelty

AMP Challenge is, to the best of our knowledge, the first competition to evaluate generative models for AMP design in a standardized and reproducible setting. The most closely related competition is the GEM workshops 2026 competition, GEM-Adaptyv, which is a RBX-1 binder design competition. However, this competition is for *de novo* protein generation against a different target.

1.3 Data

Training data. Teams are free to use any public peptide and/or AMP database for training their model. Commonly used resources such as DBAASP [20] (CC BY 4.0), APD6 [30] (unknown), dbAMP 3.0 [34] (custom, free for academic use), AMPSphere [21] (CC BY 4.0), and Peptipedia v2.0 [1] (ODbl license) provide open access to AMP data for research purposes. Notably, databases such as DBAASP not only contain the sequences but also gather minimum inhibitory concentration (MIC) values against several bacterial strains for a large portion of the sequences. Participants looking for a single consolidated resource may find the MarLys AMP database (CC-0) [15] convenient, aggregating approximately 102,000 peptide sequences from thirteen primary AMP databases.

Evaluation data. In phase 1 (computational evaluation), generated sequences are evaluated based on realism and diversity. A held-out non-public dataset is used to compute distributional similarity of the generated sequences from each team. In phase 2 (experimental evaluation), the evaluation dataset produced during the competition consists of up to 500 peptides with paired MIC values across 20 strains and HC50 values against human red blood cells. To prevent leakage of known AMPs into the top-100 candidate lists, every peptide in the top-100 must have no more than 80% sequence identity, computed via MMseqs2 [23] pairwise alignment, to any entry in the MarLys reference AMP database [15]; non-compliant candidates are replaced by the next valid entry. All submitted sequences and all experimental measurements will be released publicly under CC-BY 4.0, and all tested peptides will additionally be deposited to DBAASP [20] and APD6 [30].

Ethics. No human subjects or personally identifiable data are involved at any stage of the competition. The data collection, release, and use plan complies with the NeurIPS Code of Ethics, including provisions on privacy, consent, copyright, fair use, and representativeness.

1.4 Tasks and application scenarios

The task is *de novo generative design of linear AMPs*. Given no input beyond the competition's sequence constraints, each team submits (a) a library of 50,000 designed peptides and (b) a top-100 candidate list with documented selection procedure. The competition has five competition categories: broad-spectrum, Gram-positive, Gram-negative, multi-drug resistant (MDR), and optimal selectivity. For each category, the peptides are evaluated against a panel of bacterial strains, which can be found in Appendix B.

Relevance and difficulty. The problem setting mirrors ongoing AMP discovery efforts at academic groups including the de la Fuente lab at Penn and the Szczurek lab at Helmholtz Munich, and the bacterial panel includes all ESKAPE pathogens flagged by the WHO as priority targets. The task is challenging because antimicrobial activity depends on a subtle interplay of sequence, charge, hydrophobicity, amphipathicity, and structure, and tractable because recent generative models have demonstrated meaningful hit rates [24, 27] and any design strategy is admissible.

Ethics and dual-use. The task complies with the NeurIPS Code of Ethics. The generated data (peptide sequences, experimental measurements) are of the same kind already released through DBAASP, APD3, and dbAMP under permissive licenses. The dual-use risk surface centers on the possibility of repurposing methods toward pathogen enhancement; the competition mitigates this by restricting the task to linear peptides against well-characterized strains, excluding modifications typical of weaponizable peptide constructs, and releasing all sequences and measurements openly.

1.5 Metrics

The evaluation follows a two-phase design.

Phase 1: Computational Screening (50,000 sequences)

This phase operates on the full library of 50,000 sequences. Libraries are evaluated using the `seqme` framework [17] and the benchmark code across four metric families:

- **Surrogate activity prediction.** Aggregated scores from multiple independent AMP-classification and MIC-prediction surrogates, such as AMPredictor [4], MBC-Attention [33], DeepAMP [18] chosen to reduce any single model's bias.
- **Sequence-level metrics.** Uniqueness, internal diversity, novelty relative to known AMPs (alignment-based, using normalized bit-scores), and clustering-based coverage of the generated library.
- **Distributional similarity in embedding space.** Frechet Biological Distance [8], Maximum Mean Discrepancy [10], and precision/recall metrics [12], computed against two reference sets (a curated set of known AMPs and a generic peptide set) using ESM2 [13] and ESM-C [6] protein language model embeddings.

- **Property distribution.** Conformity [5] of charge and amphiphilicity distributions to those of known AMPs, and rate of sequences satisfying empirically derived synthesizability constraints.

Submitted sequences are also screened for exact matches against the MarLys-AMP database [15] to quantify rediscovered known peptides. This Phase 1 novelty check is distinct from the 80% identity filter applied to the top-100 list for experimental selection.

The aggregation weights, tie-breaking criteria, and curated reference-set composition used for the final ranking are held out until the close of Phase 1, to prevent direct optimization against the ranking function.

Phase 2 (experimental evaluation). From each qualifying team's top-100 list, 25 peptides are drawn uniformly at random to form a total cohort of up to 500 peptides. All peptides are synthesized by a single vendor (AAPPtec) and assayed in a single lab (de la Fuente group, University of Pennsylvania) under a standardized broth microdilution protocol with matched inoculum, media, solvent policy, and incubation conditions. The minimum inhibitory concentration (MIC), the standard measure of antimicrobial potency, is recorded in μM up to a ceiling of 64 μM (values above are reported as $> 64\,\mu$M); the half-maximal hemolytic concentration (HC50), a standard proxy for toxicity to human cells, is measured to a ceiling of 128 μM.

Scoring. Peptide-level metrics are the per-strain MIC, the MIC50 and MIC90 across the relevant strain panel, success rate (fraction of strains where MIC meets the potency threshold of 16 μM, tighter than the 32 μM cutoff used in earlier generative AMP studies [24]), HC50, and safety window $\mathrm{SW} = \mathrm{HC50}/\mathrm{MIC50}$. Team-level scores for each of the five categories are the arithmetic mean of the relevant peptide metric across the team's 25 peptides; ties are broken by MIC90 for broad-spectrum and MIC50 for the strain-specific categories.

Reproducibility. Reproducibility requirements operate at the level of submitted methods. Teams seeking co-authorship must release their generation code with a fixed random seed; organizers verify this by running `uv sync` and the entry-point script on a Linux workstation with a single GPU, comparing the regenerated library against the submitted one. Phase 1 metrics are deterministic given a fixed `seqme` version, so the relative ranking of libraries is fully reproducible from public code. Phase 2 measurements are point estimates per peptide-strain pair; team-level scores are reported alongside their constituent 25-peptide distributions.

1.6 Baselines, code, and material provided

To anchor the evaluation and give participants concrete targets, we release two baseline generators from the organizing committee's prior work:

- **HydrAMP** [24]: conditional VAE for active AMP generation.
- **AMP-Diffusion** [27]: a diffusion model operating on protein language model embeddings.

Per the organizing committee participation policy, both methods are excluded from rankings and prizes. Both methods were experimentally validated against a panel of clinically relevant strains.

At competition launch, the starter kit will additionally include:

- Trained weights and inference scripts for both methods (already publicly available).
- A 50,000-sequence library generated under default settings for each method.
- Phase 1 `seqme` [17] metrics for both baseline libraries, computed under the same protocol used for participant submissions.
- The complete set of experimentally characterized peptides from each method with their measured MIC values.

These baselines serve two functions: (1) a published Phase 1 target for participants to beat, and (2) verification that the `seqme` pipeline produces sensible relative rankings on methods with known experimental outcomes.

1.7 Accessibility and fair competition

To level the playing field for less well-resourced research groups, the competition is designed so that participation does not require substantial compute: methods running on a single CPU are eligible alongside methods using GPU-accelerated training. Any generative approach qualifies, from lightweight sampling to large-scale model-based methods; fine-tuning is not required. The most significant barrier this competition removes, however, is access to wet-lab validation: peptide synthesis and the full MIC/HC50 assay panel for qualifying teams are funded centrally by the organizers. For most academic AMP groups, this experimental component would otherwise be the limiting factor; here it is zero cost to participants.

1.8 Platform, website, and documentation

The competition will be hosted on Kaggle Community Hackathons, which will handle team registration with institutional email verification, submission management, leaderboard maintenance, and the post-submission integrity checks. The competition website, `https://szczurek-lab.github.io/amp-challenge-website/`, serves as the central information hub. The website is already live and currently includes the competition timeline, a dedicated FAQ section, step-by-step participation instructions, and the protocols for submission and evaluation. The starter kit, including the project template and the submission compliance script, is maintained at `https://github.com/szczurek-lab/amp-challenge-2027`.

1.9 Deliverables and community legacy

AMP Challenge is designed to produce community resources that outlast the competition itself. By the close of the experimental phase, the competition will release:

- An open benchmark dataset of 500 peptides with paired computational metrics and standardized MIC and HC50 measurements across the 20-strain bacterial panel, including both active and inactive sequences. To our knowledge this will be the largest such dataset spanning multiple generative methods under matched single-laboratory experimental conditions.
- A public model repository archiving the submitted methods of all co-authorship-eligible teams, with trained weights, inference code, and documentation under permissive OSI-approved licenses.
- A jointly authored benchmarking paper reporting the first controlled head-to-head comparison of generative approaches for AMP design, identifying which computational metrics and selection rules are predictive of experimental efficacy.

The evaluation infrastructure itself is released and maintained as a community resource. The `seqme` framework [17] is publicly available and will be extended with the competition's results, providing a reusable evaluation pipeline for future generative AMP methods independent of the competition cycle. All sequences and experimental measurements are released under CC-BY 4.0, and all tested peptides are additionally deposited to AMP databases, ensuring discoverability through the field's primary databases.

2 Organizational aspects

2.1 Protocol

Teams can participate in the competition in one of two ways:

Minimum requirements (benchmark participation). All teams must provide the following to be included in the experimental benchmark:

- A short abstract summarizing the method.
- A generated peptide library of 50,000 designed AMPs for computational evaluation.
- A ranked candidate list of the top 100 AMPs, with documentation of the selection procedure.

- A brief description of training data, external databases used, and any manual interventions or computational filters applied.

Full requirements (co-authorship eligibility). To be eligible for co-authorship on the benchmarking paper, teams must additionally provide:

- A public GitHub repository following the provided template (`https://github.com/szczurek-lab/amp-challenge-2027`), containing trained model weights, inference code, and detailed usage documentation.
- A permissive OSI-approved license (e.g., MIT, BSD-3-Clause, or Apache 2.0) specified in the repository.
- A fixed default random seed such that running the generation script twice produces identical output.
- Full training data disclosure. If any proprietary or non-public data was used, it must be released publicly under a permissive license as part of the submission. Teams should verify that their use of any data source complies with that source's terms of use.

Organizers will verify reproducibility by running `uv sync` followed by the designated entry-point script, and comparing the output against the submitted library. Teams that satisfy only the minimum requirements will receive their experimental results and be included in the benchmark, but will not be eligible for co-authorship.

Qualification for experimental validation. Should the number of qualifying teams exceed experimental capacity, teams will be ranked by computational evaluation of their peptide library and top candidate list. A maximum of 20 teams will advance to experimental validation.

Pre-launch validation. Before the June 2026 full launch, organizers dry-run the full submission and evaluation pipeline using HydrAMP and AMP-Diffusion as synthetic submissions, covering compliance checks, `seqme` evaluation, the reproducibility verification flow, and the integrity checks of Section 2.3.

2.2 Rules and Engagement

The following rules govern participation in AMP Challenge and are reproduced verbatim on the competition website.

Eligibility. AMP Challenge is open to any team worldwide, including academic groups, industry researchers, and independent researchers. There is no registration fee. Teams may consist of any number of members. Each team may submit only one set of outputs (one 50,000-peptide library and one top-100 list). A participant may contribute to more than one team only if the methods are substantively different (distinct architecture, training data, or design principle); near-duplicate submissions across nominally different teams are not permitted and will result in disqualification, as detailed in Section 2.3. Organizing Committee members may participate scientifically but are ineligible for prizes and rankings; their peptides are tested in a separate experimental batch and do not count toward the 500-peptide budget allocated to competing teams.

Submission format and compliance. All submissions must be provided in the format specified in the starting kit repository. Before formal evaluation, every submission undergoes an automated compliance check verifying: sequence validity (standard amino acid alphabet); length bounds (8–50 residues); uniqueness within the library; the linear-peptide requirement; and metadata completeness. Teams failing the automated check are notified and given a 72-hour window to resolve issues before disqualification.

Data and model use. Teams may train on any publicly available peptide database. Use of proprietary or non-public data is permitted provided the data are released publicly under a permissive license at the time of submission. Teams must disclose all training data sources; misrepresentation of data provenance will result in disqualification.

Communication channels. All official communications (timeline changes, rule clarifications, results) are sent by email to registered team contacts and posted on the competition website. Technical questions may be submitted via GitHub Issues on the competition repository (`https://github.com/szczurek-lab/amp-challenge-2027`). Non-technical questions may be directed to the organizer email address.

2.3 Anti-collusion and submission integrity

The competition adheres to the anti-collusion policies of the NeurIPS Main Track Handbook. To prevent gaming of the experimental budget through duplicate registrations, and to maintain leaderboard integrity, the following measures are in place.

Registration. Each team must register with an institutional email address (university, research institute, or company domain). Student email addresses from accredited institutions are accepted on the same basis as faculty addresses. Free webmail accounts (gmail, hotmail, yahoo, and equivalents) are not accepted for primary registration. Participants may contribute to multiple teams provided each team's submission represents a genuinely distinct method (different architecture, training data, or design principle). Multi-team participation should be declared at registration. All such submissions are subject to the same post-submission integrity checks as unrelated teams.

Post-submission integrity checks. After submissions close, organizers perform two automated checks across all submitted libraries:

- **Sequence overlap analysis.** Pairwise comparison of submitted 50,000-sequence libraries and top-100 candidate lists, flagging team pairs with substantially overlapping sequence sets as evidence of collusion or shared submissions.
- **Method-description similarity.** Pairwise comparison of submitted method abstracts and documentation.

Teams flagged by either check are reviewed by the organizing committee. Confirmed cases of collusion or duplicate-team submissions may result in disqualification of affected teams from the experimental phase and from co-authorship eligibility on the benchmarking paper.

Use of AI assistants. Per the NeurIPS Main Track Handbook, the use of AI assistants and large language models in preparing submissions is permitted, but must be disclosed in the method documentation. Human team members remain responsible for the submitted content.

2.4 Schedule and readiness

- **April 2026:** Pre-competition call opens; early registration portal and website go live.
- **June 2026:** Full competition call and registration open.
- **October 1st 2026 AOE:** Submission deadline (50,000-peptide library and top-100 list).
- **October 2026:** Phase 1 computational evaluation; qualification decisions communicated to teams.
- **December 2026:** NeurIPS workshop; announcement of computational evaluation winners and teams advancing to experimental validation.

Post-workshop experimental follow-up and paper writing:

- **October 2026 – February 2027:** Peptide synthesis window (AAPPtec vendor).
- **March – April 2027:** Experimental validation (MIC/HC50 assays) and final data analysis (de la Fuente group, University of Pennsylvania).
- **TBA 2027:** Experimental results shared at a dedicated event.
- **Late 2027:** Benchmarking paper submission.

Readiness. The competition website is already live, the starting kit repository is publicly available, and the wet-lab protocol has been finalized with the de la Fuente group. The bacterial panel and synthesis vendor (AAPPtec) have been confirmed.

Contingency plan. If fewer than 20 teams qualify, unused experimental capacity is redistributed as additional peptides per qualifying team. If more than 20 qualify, the Phase 1 ranking selects the advancing cohort. Vendor-side synthesis delays can extend the 2027 experimental window by up to two months without affecting the workshop, which presents Phase 1 results rather than experimental outcomes. Platform outages around the deadline trigger a Google Forms backup endpoint with manually applied compliance checks.

2.5 Competition promotion and incentives

The competition is promoted through our active accounts on X, LinkedIn, and BlueSky, and directly via email outreach to research labs working on generative modeling and AMP design. We are additionally committed to targeted outreach to groups in Low- and Middle-Income Countries (LMICs), particularly in regions with a high antimicrobial resistance burden such as South and Southeast Asia and sub-Saharan Africa, where the clinical need for new antimicrobials is most acute. Outreach includes direct dissemination of the call through international AMR-focused networks and consortia (e.g., the Global AMR R&D Hub and WHO regional AMR programmes) and email contact with corresponding authors of recent AMP publications from research groups in target regions.

The primary award for qualifying teams is the funding of peptide synthesis and experimental validation: 25 peptides per team, across up to 20 qualifying teams, synthesized and assayed at no cost to participants. Additionally, teams meeting all full requirements are eligible for co-authorship on the benchmarking paper. We are also reaching out to industry sponsors to offer cash prizes for the top-ranked teams.

2.6 Competition track workshop and dissemination

We request a two-hour slot at the NeurIPS 2026 Competition Track in-person event in December 2026. The most important function of the workshop is to publicly announce the up to 20 teams whose peptides advance to experimental validation in Phase 2, alongside the Phase 1 evaluation results across all submitted libraries. The session is structured as approximately 30 minutes of organizer-led presentation (competition design, Phase 1 results, qualifying teams, the plan for the benchmarking paper, and award announcements), followed by short talks from selected qualifying teams, a poster session for all co-authorship-eligible teams, and a closing panel.

3 Resources

3.1 Organizing team

The organizing team spans two continents and combines deep expertise in generative modeling of biological sequences (Szczurek lab, Helmholtz Munich and University of Warsaw), state-of-the-art AMP wet-lab validation (de la Fuente group, University of Pennsylvania), and competition logistics and reproducibility infrastructure. Brief biographies are provided in Appendix A. The organizing team includes researchers at multiple career stages (PhD students and senior faculty), spans three countries (Germany, Poland, USA), and includes gender diversity among both junior and senior members.

3.2 Resources provided by organizers

The organizing labs cover all peptide synthesis (Szczurek lab, Helmholtz Munich and University of Warsaw) and wet-lab validation (de la Fuente group, University of Pennsylvania); no external compute, equipment, or sponsorship is required from the conference.

3.3 Support requested

We request a two-hour slot at the NeurIPS 2026 Competition Track in-person event, with venue suitable for an organizer presentation and a poster session, and conference-side logistical support (scheduling, AV, room turnover). The session content is described in Section 2.6.

A Biography of all team members

Paulina Szymczak paulina.szymczak@helmholtz-munich.de

Paulina Szymczak is a PhD candidate at the Institute of AI for Health, Helmholtz Munich, and at LMU Munich, supervised by Ewa Szczurek. Her doctoral research, within the ERC-funded DOG-AMP project, focuses on generative AI for AMP discovery and on coupling generative methods with wet-lab validation. She is first author on HydrAMP [24], a conditional VAE for active AMP generation experimentally validated against multi-drug resistant strains, and a co-author on OmegAMP [22], a targeted AMP discovery framework, and on a recent survey of AI-driven AMP discovery covering both mining and generation [25]. She was previously a visiting researcher at Northwestern University Feinberg School of Medicine. She holds an MSc in Bioinformatics and Systems Biology from the University of Warsaw and a BSc in Biotechnology from the University of Wrocław.

Rasmus Møller-Larsen rasmus.larsen@helmholtz-munich.de

Rasmus Møller-Larsen is a PhD candidate at the Institute of AI for Health, Helmholtz Munich, and at the Technical University of Munich, supervised by Ewa Szczurek and Niki Kilbertus. His doctoral research, within the ERC-funded DOG-AMP project, focuses on generative AI for multi-objective optimization – with a focus on antimicrobial peptide discovery. He is the first author on `seqme` [17], which is a computational framework to evaluate biological sequence designs from generative model.

Bruno Puczko-Szymański b.puczko-szyma@student.uw.edu.pl

Bruno Puczko-Szymański is a MSc graduate in Bioinformatics from the Faculty of Mathematics, Mechanics and Informatics at the University of Warsaw. His master's thesis, titled "Evaluating Metrics for Generative Models in Antimicrobial Peptide Design", focused on the assessment and benchmarking of evaluation metrics for generative models applied to antimicrobial peptide discovery. His relevant experience includes participation in research projects related to the evaluation and design of AMP [17].

Marcelo Der Torossian Torres marcelot@pennmedicine.upenn.edu

Marcelo Der Torossian Torres is a Research Associate at the University of Pennsylvania, working under the mentorship of Cesar de la Fuente in the Departments of Bioengineering and Microbiology. His research focuses on integrating experimental and computational approaches for peptide discovery, with particular emphasis on AI-driven design of antimicrobial agents, nanomaterials, and biosensors. His relevant work includes the design and synthesis of diverse bioactive molecules, ranging from linear peptides to complex constrained analogs, leading to the development of selective antiprotozoal compounds and engineered toxin-derived therapeutics [26, 27].

Cesar de la Fuente cfuente@pennmedicine.upenn.edu

César de la Fuente is a Presidential Associate Professor at the University of Pennsylvania, where he leads the Machine Biology Group. He earned his PhD from the University of British Columbia and completed postdoctoral research at MIT. His lab pioneered computational and AI approaches to antibiotic discovery that have accelerated the identification of preclinical antimicrobial candidates from years to hours, with a sustained track record of coupling generative design with experimental validation against clinically relevant pathogens. He is senior author on AMP-Diffusion [27], one of the two baseline methods in this competition, and co-authored a recent survey of AI-driven AMP discovery [25]. His lab also originated the concept of molecular de-extinction, recovering therapeutic peptide sequences from extinct organisms, and has uncovered novel antimicrobial molecules across the tree of life [29]. He is a recipient of the Fleming Prize, recognizing outstanding contributions to antimicrobial research. In this competition, his group leads the standardized wet-lab experimental validation pipeline, including MIC and HC50 assays against the 20-strain bacterial panel.

Ewa Szczurek ewa.szczurek@helmholtz-munich.de

Ewa Szczurek is an associate professor at the Faculty of Mathematics, Informatics and Mechanics at the University of Warsaw, Poland and serves as the director of the Institute of AI for Health at Helmholtz Munich, Germany. She leads joint labs at Helmholtz Munich and at the University of Warsaw. She was visiting associate professor at Northwestern University in the United States

(2023) and a visiting fellow at the Center for Interdisciplinary Research, Bielefeld, Germany (2016). She holds Master degrees in computer science from the University of Warsaw, Poland and Uppsala University, Sweden. She obtained her doctoral degree from the Max Planck Institute for Molecular Genetics in Berlin (2011), followed by a postdoctoral fellowship in Switzerland at ETH Zurich. She received the distinction, scientific and didactic awards from the Rector of the University of Warsaw, as well as ETH Zurich, IMPRS fellowships for her postdoctoral and doctoral research and scientific excellence award from Perspektywy Foundation and was distinguished as Forbes Woman to Watch 2026. Her work was supported by numerous grants including ERC consolidator. She acts as area chair for NeurIPS and ICML, and a program committee member for the ISMB and RECOMB-CCB conferences, as well as associate editor for Genome Biology. She loves to work on AI models and is very excited about their applications in molecular biology and medicine [16, 17, 24, 25].

B Bacterial strains

Peptides will be evaluated on a panel of 20 bacterial strains. Multi drug resistant strains are marked with [MDR] below.

Gram-negative (15 strains)

1. *A. baumannii* ATCC 19606
2. *A. baumannii* ATCC BAA-1605 (CGTPACCIMRA) [MDR]
3. *E. cloacae* ATCC 13047
4. *E. coli* ATCC 11775
5. *E. coli* AIC221
6. *E. coli* AIC222 (CRE) [MDR]
7. *E. coli* ATCC BAA-3170 (CRE) [MDR]
8. *E. coli* K-12 BW25113
9. *K. pneumoniae* ATCC 13883
10. *K. pneumoniae* ATCC BAA-2342 (EIRK) [MDR]
11. *P. aeruginosa* PAO1
12. *P. aeruginosa* PA14
13. *P. aeruginosa* ATCC BAA-3197 (FBCRP) [MDR]
14. *S. enterica* ATCC 9150
15. *S. enterica* Typhimurium ATCC 700720

Gram-positive (5 strains)

1. *B. subtilis* ATCC 23857
2. *S. aureus* ATCC 12600
3. *S. aureus* ATCC BAA-1556 (MRSA) [MDR]
4. *E. faecalis* ATCC 700802 (VRE) [MDR]
5. *E. faecium* ATCC 700221 (VRE) [MDR]

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